

The empirical recurrence for OEIS A237205: recovering a conjecture that is not written down

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Abstract

OEIS A237205 records “Empirical recurrence of order 97”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 260$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 97, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 97 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237205 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the upper median plus the lower median minus the minimum of every 2×2 subblock equal”. It has offset 1 and begins

1160, 8396, 59386, 472096, 3795060, 31758656, 268073808, 2299621682, ...

The entry states:

Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:57:15, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 260$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 260$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 520$ exact terms of the model returns order 97 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	48	c_{26}	−3704534786993514968717	c_{51}	4654578473281696701345761562
c_2	−725	c_{27}	−23309644238280295350506	c_{52}	4026519979135276271100816776
c_3	−1397	c_{28}	60495635056120042040895	c_{53}	−101560055957586365875684783
c_4	154573	c_{29}	194822932655084744007524	c_{54}	−665700402641785146537027191
c_5	−1203582	c_{30}	−722486090437205379100869	c_{55}	1908891750401356052946149380
c_6	−9125921	c_{31}	−1268520318721766147911037	c_{56}	9487930509859026078105013093
c_7	172918023	c_{32}	6838176599076022124741273	c_{57}	−3095254024510918795652058507
c_8	−183811339	c_{33}	5962970057441652507949990	c_{58}	−116526350676387979929064580
c_9	−11134754234	c_{34}	−53117213260041016030654770	c_{59}	4330441304503591985619121911
c_{10}	56334971836	c_{35}	−14022291182065497148269244	c_{60}	1234064140144278056894630149
c_{11}	382772090568	c_{36}	345112973568181199691433290	c_{61}	−522241725369792232556583337
c_{12}	−3849058875490	c_{37}	−66546807921040156041349292	c_{62}	−113042738942037902964559906
c_{13}	−4712882843244	c_{38}	−1897479349505260940697461177	c_{63}	5418265486816792628141995827
c_{14}	154812445590369	c_{39}	1062451081445559886781251783	c_{64}	9020638482944390254594295215
c_{15}	−208982516000771	c_{40}	8896087906548502767506124765	c_{65}	−482191073652635486699912962
c_{16}	−4216005124250115	c_{41}	−7758802196894948626439608785	c_{66}	−635063124714120999767672482
c_{17}	14007280596139027	c_{42}	−35749020908143410573655203943	c_{67}	3666333949268977531992059945
c_{18}	79870011611456699	c_{43}	41170331916941327130500454839	c_{68}	4012979547537771118107425766
c_{19}	−451200046046428157	c_{44}	123558151314061576338306950065	c_{69}	−236980228503849646594164853
c_{20}	−980534253355536719	c_{45}	−174716858407169818827940388907	c_{70}	−231172202645324134844495031
c_{21}	10029112048539086465	c_{46}	−368108124068077094817065522420	c_{71}	1294037430109688421551866429
c_{22}	4306497960432349170	c_{47}	615354413737812111560623862091	c_{72}	1217777115148815011704206965
c_{23}	−168493143568018209047	c_{48}	946467480415475809592843001210	c_{73}	−592402101070861815491658633
c_{24}	122424238017993871871	c_{49}	−1831694234176147913353637518723	c_{74}	−577323296014175754319058378
c_{25}	2220553839965369099423	c_{50}	−2101118675728876324255185956526	c_{75}	2252581358745338383240331892

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 97, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $97 + 4$ terms removed returns order 97 as well, so $\deg q_{\min} = 97 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $97 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 97 that this sequence satisfies — and that family turns out to have exactly one member.

References

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